

C03-AT3

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Data Structures
CSA0312

1. Construct the Binary Tree

pre-order : 1, 2, 4, 5, 3, 6, 7

In-order : 4, 2, 5, 1, 6, 3, 7

In preorder, the first element is the root.

Root = 1

In inorder, left subtree = 4, 2, 5

right subtree = 6, 3, 7

left subtree

Preorder for left = 2, 4, 5

Root = 2

Inorder : 4, 2, 5

So, left child = 4

right child = 5

Right subtree

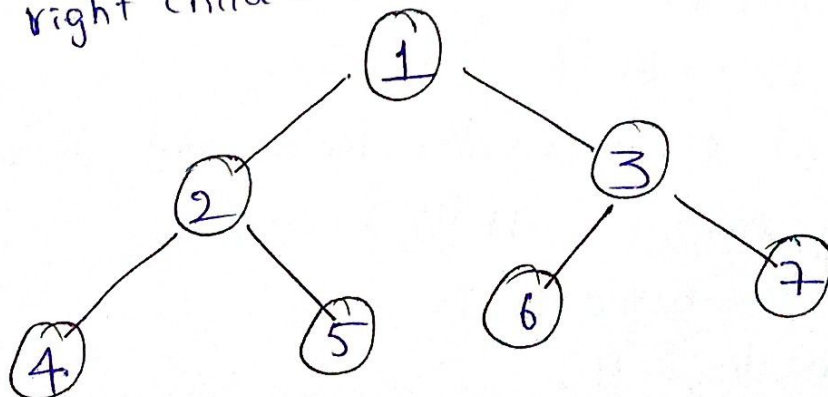
Preorder for right = 3, 6, 7

Root = 3

Inorder : 6, 3, 7

So, left child = 6

right child = 7



Preorder :

1 $\rightarrow$ 2 $\rightarrow$ 4 $\rightarrow$ 5 $\rightarrow$ 3 $\rightarrow$ 6 $\rightarrow$ 7

Inorder :

4 $\rightarrow$ 2 $\rightarrow$ 5 $\rightarrow$ 1 $\rightarrow$ 6 $\rightarrow$ 3 $\rightarrow$ 7

2. Construct Binary Tree In-order : H, D, I, B, E, A, F, C, G
post-order : H, I, D, E, B, F, G, C, A

In postorder, the last element is the root

Root = A

Now locate A in inorder : H D I B E A F C G

Therefore, left subtree

Inorder : H, D, I, B, E

postorder : H, I, D, E, B

Right subtree

Inorder : F, C, G

postorder : F, G, C

left subtree

last element of postorder is the root, Root = B

locate B in inorder : H D I B E

Therefore, left subtree of B

Inorder : H, D, I

postorder : H, I, D

Right subtree of B

Inorder : E

postorder : E

last element of postorder is the root, Root = D

locate D in inorder : H D I

Therefore, left subtree of D

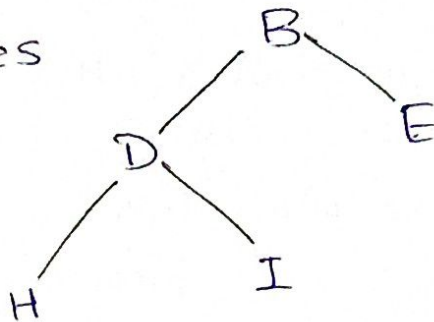
Inorder : H

postorder : H

So, H is the left child of D.
Right subtree of D - Inorder : I
postorder : I

So, I is the right child of D.

Left subtree becomes



Right subtree

last element of postorder is the root.

Root = C

locate C in inorder : F C G

therefore, left subtree of C

Inorder : F

postorder : F

So, F is the child of C.

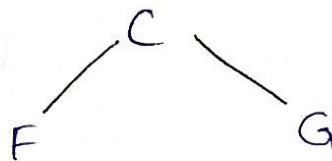
Right subtree of C

Inorder : G

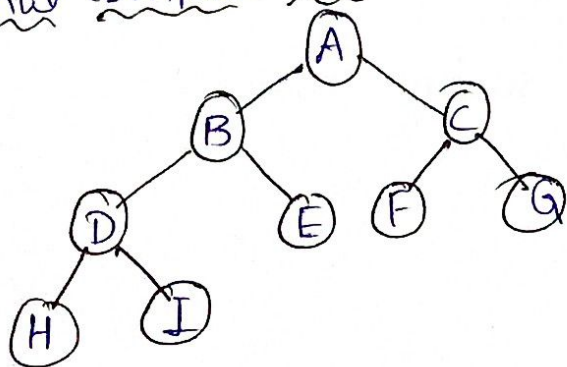
postorder : G

So, G is the right child of C.

Right subtree becomes



Final Complete Tree



Preorder rule :- Root $\rightarrow$ Left $\rightarrow$ Right

first root : A

left subtree : B, D, H, I, E

right subtree : C, F, G

Therefore, preorder = A, B, D, H, I, E, C, F, G

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Construct the tree using Inorder : D, B, E, A, F, C, G

preorder : A, B, D, E, C, F, G

In preorder, the first element is always the root.

Root = A.

Locate A in inorder : D B E A F C G

So, left subtree = D, B, E

right subtree = F, C, G

left subtree

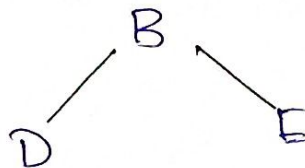
Remaining preorder : B, D, E

first element is B. so, left subtree root = B

In-order : D B E

therefore, left child of B = D

right child of B = E



Right subtree

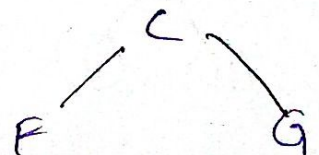
Remaining preorder : C, F, G

first element is C. so, right subtree root = C

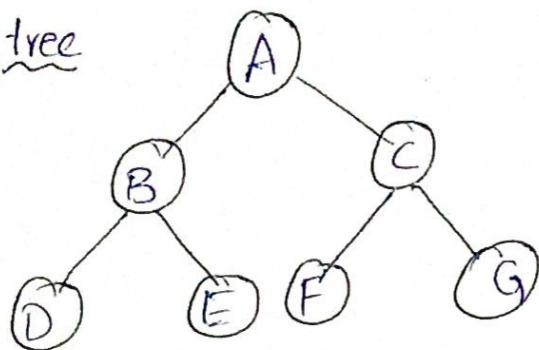
In-order : F C G

therefore, left child of C = F

right child of C = G



Final tree



Postorder Traversal

postorder rule : Left $\rightarrow$ Right $\rightarrow$ Root

left subtree : D, E, B

right subtree : F, G, C

Root : A

therefore, Postorder = D, E, B, F, G, C, A

4. Insert the following elements in a BST

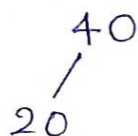
40, 20, 60, 10, 30, 50, 70, 25, 35

Insert 40

40 becomes the root

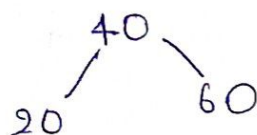
Insert 20

$20 < 40$



Insert 60

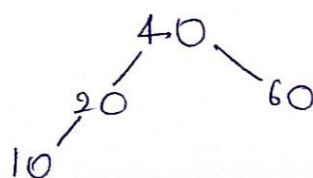
$60 > 40$



Insert 10

$10 < 40$

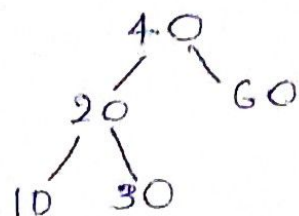
$10 < 20$



Insert 30

$$30 < 40$$

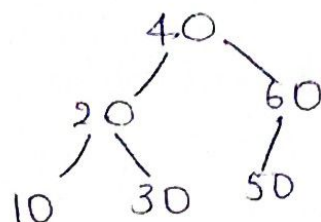
$$30 > 20$$



Insert 50

$$50 > 40$$

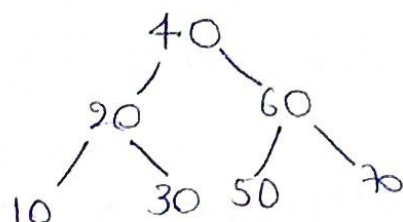
$$50 < 60$$



Insert 70

$$70 > 40$$

$$70 > 60$$

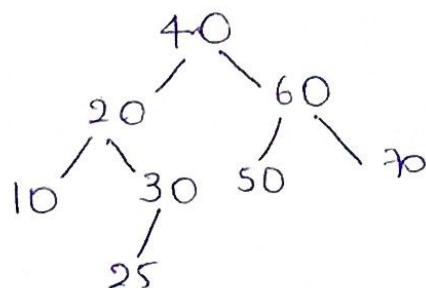


Insert 25

$$25 < 40$$

$$25 < 20$$

$$25 < 30$$

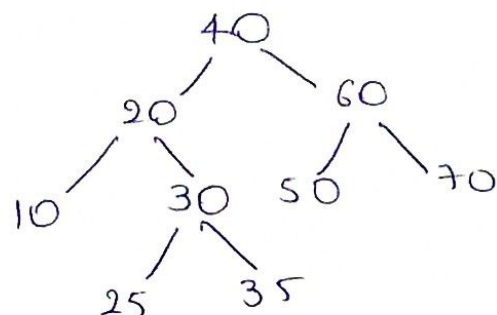


Insert 35

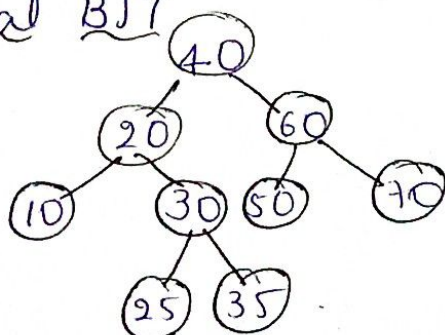
$$35 < 40$$

$$35 > 20$$

$$35 > 30$$



Final BJT



Preorder

Root $\rightarrow$ Left $\rightarrow$ Right

40, 20, 10, 30, 25, 35, 60, 50, 70

Inorder

Left $\rightarrow$ Root $\rightarrow$ Right

10, 20, 25, 30, 35, 40, 50, 60, 70

Level order

Level 1 : 40

Level 2 : 20, 60

Level 3 : 10, 30, 50, 70

Level 4 : 25, 35

level order = 40, 20, 60, 10, 30, 50, 70, 25, 35

Post order

Left $\rightarrow$ Right $\rightarrow$ Root

10, 25, 35, 30, 20, 50, 70, 60, 40

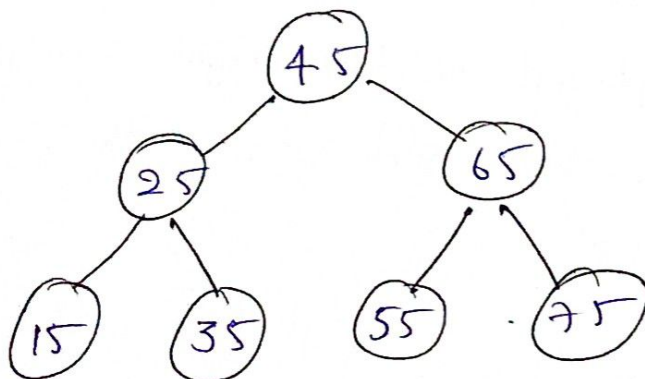
5. Insert the following 45, 25, 15, 35, 65, 55, 75

$\rightarrow$ Preorder

$\rightarrow$ Postorder

$\rightarrow$ Level order

Insertion order :- 45, 25, 15, 35, 65, 55, 75



TRAVERSALS

Preorder : Root $\rightarrow$ Left $\rightarrow$ Right

Root (first element) = 45

left subtree : 25, 15, 35

right subtree : 65, 55, 75

$\therefore$ Preorder = 45, 25, 15, 35, 65, 55, 75

Postorder : Left $\rightarrow$ Right $\rightarrow$ Root

Root = 45

left subtree : 15, 35, 25

right subtree : 55, 75, 65

$\therefore$ Postorder = 15, 35, 25, 55, 75, 65, 45

Level order : Visit the tree by level

Level 1 : 45

Level 2 : 25, 65

Level 3 : 15, 35, 55, 75

$\therefore$ level order = 45, 25, 65, 15, 35, 55, 75

Inorder : Left $\rightarrow$ Root $\rightarrow$ Right

left subtree : 15, 25, 35

Root = 45

right subtree : 55, 65, 75

$\therefore$ Inorder = 15, 25, 35, 45, 55, 65, 75

Justification

$\Rightarrow$ BST property is followed: smaller elements are placed in the left subtree and the larger elements are placed in the right subtree.

Traversals are obtained using their respective rules.